TABLE 8
Representative spatial & temporal attentions sorted by date. Action=action recognition, ReID = re-identification. Ranges means the ranges of attention map. S or H means soft or hard attention. $g(x)$ and $f(g(x), x)$ are the attention process described by Eq. 1. (A)element-wise product.(B) aggregate information via attention map.(I) emphasize key points in both spatial and temporal domains, (II)capture global information.

Category	Method	Publication	Tasks	$g(x)$	$f(g(x), x)$	Ranges	S or H	Goals
Separately predict spatial & temporal attention	STA-LSTM [130]	AAAI2017	Action	a)spatial: fuse hidden state -> MLP -> Softmax, b)temporal: fuse hidden state -> MLP -> ReLU	(A)	(0,1), (0,+∞)	S	(I)
	RSTAN [16]	TIP2018	Action	a)spatial: fuse hidden state -> MLP -> Softmax, b)temporal: fuse hidden state -> MLP -> Softmax	(B)	(0,1)	S	(I) (II)
Jointly predict spatial & temporal attention	STA [131]	AAAI2019	ReID	a) tenporal: produce per-frame attention maps using l_2 norm b) spatial: obtain spatial scores for each patch by summation using l_1 norm.	(B)	(0,1)	S	(I)
Pairwise relation-based method	STGCN [177]	CVPR2020	ReID	construct a patch graph using pairwise similarity	(B)	(0,1)	S	(I)

Finally, STA splits the input feature map X_i into K regions $\{X_{n,1}, \ldots, X_{n,K}\}$ and computes the output using

$$Y^1 = [X_{\arg\max_n S(n,1),1}; \ldots; X_{\arg\max_n S(n,K),K}] \tag{161}$$

$$Y^2 = [\sum_{n=1}^{N} S(n,1)X_{n,1}; \ldots; \sum_{n=1}^{N} S(n,K)X_{n,K}] \tag{162}$$

$$Y = [Y^1; Y^2] \tag{163}$$

Instead of computing spatial attention maps frame by frame, STA considers spatial and temporal attention information simultaneously, fully using the discriminative parts in both dimensions. This reduces the influence of occlusion. Because of its non-parametric design, STA can tackle input video sequences of variable length; it can be combined with any 2D CNN backbone.

3.7.4 STGCN

To model the spatial relations within a frame and temporal relations across frames, Yang et al. [177] proposed a novel *spatiotemporal graph convolutional network* (STGCN) to learn a discriminative descriptor for a video. It constructs a patch graph using pairwise similarity, and then uses graph convolution to aggregate information.

STGCN includes two parallel GCN branches, the temporal graph module and the structural graph module. Given the feature maps of a video, STGCN first horizontally partitions each frame into P patches and applies average pooling to generate patch-wise features $x_1, \ldots, x_N$, where the total number of patches is $N = TP$. For the temporal module, it takes each patch as a graph node and construct a patch graph for the video, where the adjacency matrix $\widehat{A}$ is obtained by normalizing the pairwise relation matrix E, defined as

$$E(i,j) = (W^{\phi} x_i)^T W^{\phi} x_j \tag{164}$$

$$A(i,j) = E^2(i,j) / \sum_{j=1}^{N} E^2(i,j) \tag{165}$$

$$\widehat{A} = D^{-\frac{1}{2}}(A+I)D^{-\frac{1}{2}} \tag{166}$$

where $D(i,i) = \sum_{j=1}^{N}(A+I)(i,j)$. Given the adjacency matrix $\widehat{A}$, the m-th graph convolution can be found using

$$X^m = \widehat{A} X^{m-1} W^m + X^{m-1} \tag{167}$$

where $X \in \mathbb{R}^{N \times c}$ represents the hidden features for all patches and $W^m \in \mathbb{R}^{c \times c}$ denotes the learnable weight matrix for the m-th layer. For the spatial module, STGCN follows a similar approach of adjacency matrix and graph convolution, except for modeling the spatial relations of different regions within a frame.

Flattening spatial and temporal dimensions into a sequence, STGCN applies the GCN to capture the spatiotemporal relationships of patches across different frames. Pairwise attention is used to obtain the weighted adjacency matrix. By leveraging spatial and temporal relationships between patches, STGCN overcomes the occlusion problem while also enhancing informative features. It can used with any CNN backbone to process video.

4 Future Directions

We present our thoughts on potential future research directions.

4.1 Necessary and sufficient condition for attention

We find the Eq. 1 is a necessary condition but not a necessary and sufficient condition. For instance, GoogleNet [178] conforms to the above formula, but does not belong to the attention mechanisms. Unfortunately, we find it difficult to find a necessary and sufficient condition for all attention mechanisms. The necessary and sufficient conditions for the attention mechanism are still worth exploring which can promote our understanding of attention mechanisms.

4.2 General attention block

At present, a special attention mechanism needs to be designed for each different task, which requires considerable effort to explore potential attention methods. For instance, channel attention is a good choice for image classification,

while spatial attention is well-suited to dense prediction tasks such as semantic segmentation and object detection. Channel attention focuses on *what to pay attention to* while spatial attention considers *where to pay attention*. Based on this observation, we encourage consideration as to whether there could be a general attention block that takes advantage of all kinds of attention mechanisms. For example, a soft selection mechanism (branch attention) could choose between channel attention, spatial attention and temporal attention according to the specific task undertaken.

4.3 Characterisation and interpretability

Attention mechanisms are motivated by the human visual system and are a step towards the goal of building an interpretable computer vision system. Typically, attention-based models are understood by rendering attention maps, as in Fig. 9. However, this can only give an intuitive feel for what is happening, rather than precise understanding. However, applications in which security or safety are important, such as medical diagnostics and automated driving systems, often have stricter requirements. Better characterisation of how methods work, including modes of failure, is needed in such areas. Developing characterisable and interpretable attention models could make them more widely applicable.

4.4 Sparse activation

We visualize some attention map and obtains consistent conclusion with ViT [34] shown in Fig. 9 that attention mechanisms can produce sparse activation. There phenomenon give us a inspiration that sparse activation can achieve a strong performance in deep neural networks. It is worth noting that sparse activation is similar with human cognition. Those motivate us to explore which kind of architecture can simulate human visual system.

4.5 Attention-based pre-trained models

Large-scale attention-based pre-trained models have had great success in natural language processing [85], [179]. Recently, MoCoV3 [84], DINO [180], BEiT [85] and MAE [169] have demonstrated that attention-based models are also well suited to visual tasks. Due to their ability to adapt to varying inputs, attention-based models can deal with unseen objects and are naturally suited to transferring pretrained weights to a variety of tasks. We believe that the combination of pre-training and attention models should be further explored: training approach, model structures, pre-training tasks and the scale of data are all worth investigating.

4.6 Optimization

SGD [181] and Adam [182] are well-suited for optimizing convolutional neural networks. For visual transformers, AdamW [183] works better. Recently, Chen et al. [184] significantly improved visual transformers by using a novel optimizer, the *sharpness-aware minimizer* (SAM) [185]. It is clear that attention-based networks and convolutional neural networks are different models; different optimization methods may work better for different models. Investigating new optimzation methods for attention models is likely to be worthwhile.

4.7 Deployment

Convolutional neural networks have a simple, uniform structure which makes them easy to deploy on various hardware devices. However, it is difficult to optimize complex and varied attention-based models on edge devices. Nevertheless, experiments in [46], [47], [48] show that attention-based models provide better results than convolutional neural networks, so it is worth trying to find simple, efficient and effective attention-based models which can be widely deployed.

5 Conclusions

Attention mechanisms have become an indispensable technique in the field of computer vision in the era of deep learning. This survey has systematically reviewed and summarized attention mechanisms for deep neural networks in computer vision. We have grouped different attention methods according to their domain of operation, rather than by application task, and show that attention models can be regarded as an independent topic in their own right. We have concluded with some potential directions for future research. We hope that this work will encourage a variety of potential application developers to put attention mechanisms to use to improve their deep learning results. We also hope that this survey will give researchers a deeper understanding of various attention mechanisms and the relationships between them, as a springboard for future research.

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